AutoCAD Electrical 2025: AutoLISP Reference

Section E - Drawing Setup

Topics Send Feedback

Description

Insert and edit ladders. Drawing setup. Title block.

Topics

Name	Description
c:ace_find_wdt	Find the ".wdt" title block mapping file to use for project.
c:ace_GBL_wd_m	Read WD_M block on active drawing and fill and return GBL_wd_ global.
c:ace_ladr2_reseq_main	Project-wide ladder reference renumber.
c:wd_1_reset_dwg	Reset AcadE drawing-specific global variables. For example, whe different drawing is made active.
c:wd_chg_ldr_info	Retrieve list of ladder parameters from current drawing.
c:wd_get_gridrefs	Retrieve list of ladder parameters from current drawing.
c:wd_in_ladder	Insert ladder.
c:wd_ladr_reref	Freshen ladder reference numbers on active drawing if they inclu SHEET attribute and the WD_M block's SHEETvalue differs from t on each ladder's master line reference block insert.
c:wd_prnt_time_date	Add current time and date stamp to title block. Annotate title blo with current time and date stamp information if target attributes present.
c:wd_read_dwg_params	Read AutoCAD Electrical's WD_M block for active drawing and fill internal global GBL_wd_m. Insert the WD_M block if it is not fou the active drawing (user is prompted for permission to insert).
c:wd_ref_col	Re-reference ladder reference numbers on active drawing.
c:wd_renum_ladders	Re-reference ladder reference numbers on active drawing with minimum information required.
c:wd_reread_dwg_params	Provided for backwards compatibility. Use wd_read_dwg_params ace_GBL_wd_m instead.
c:wd_tb_process_one	Update title block attribute values.

Links

Topics

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Description

Find the ".wdt" title block mapping file to use for project.

AutoLISP

```
(c:ace_find_wdt wdpfnam msgs)
```

Parameters

Parameters	Description
wdpfnam	full path/filename of project ".wdp" or nil=active pro
msgs	1 = show error messages on the command line, nil = suppress error messages.

Example

Find the ".wdt" file to use for the active project.

```
(setq wdtfnam (c:ace_find_wdt nil nil))
```

Module

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c:ace_GBL_wd_m

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Description

Read WD_M block on active drawing and fill and return GBL_wd_m global.

AutoLISP

```
(c:ace_GBL_wd_m options)
```

Parameters	Description
options	1's bit set = insert WD_M block if it does not exist w any user prompt. 1's bit clear = prompt user for permission to insert V block when it does not exist on the active drawing. 2's bit set = if WD_M block does not exist then temporarily insert a copy, read to fill GBL_wd_m, the immediately delete the inserted instance and purge i

Returns

Copy of global list GBL_wd_m (or nil if failure) where:

nth 0 = rungs Vert or Horiz (V/H).

nth 1 = Ref numbering system 4=X,Y grid; 1=lref txt with bar, 2=txt only; 3=user lref symbols used.

nth 2 = TAGMODE value ("R" or "S").

nth 3 = TAG-START value (ex: "1").

nth 4 = TAG-RSUF value (ex: ",A,B,C,D,E,F").

nth 5 = TAGFMT value (ex: "%f %n").

nth 6 = WIREMODE value.

nth 7 = WIRE-START string value (ex: "100").

nth 8 = WIRE-RSUF value (ex: ",A,B,C,D,E,F,G,H,J").

nth 9 = WIREFMT value (ex: "%n").

nth 10 = WINC wire number increment.

nth 11 = RUNGDIST default rung spacing.

nth 12 = DLADW default ladder width.

nth 13 = SHEET attribute text.

nth 14 = XREFFMT cross ref format (ex: "%s-%n").

nth 15 = RUNGINC rung increment (default = 1).

nth 16 = DRWRUNG (= 0 don't drawing ladder rungs, 1 = draw all, 2 = every other, etc.).

nth 17 = PH3SPACE 3-phase bus spacing value.

nth 18,19 = DATUMX,DATUMY (X-Y grid and X-zone X and Y origin values).

nth 20,21 = DISTH,DISTV (X-Y grid and X-zone increment widths).

nth 22,23 = CHAR_H,CHAR_V (X-Y grid and X-zone label list).

nth 24 = X-Y horiz first flag (X-Y grid format).

nth 25 = X-Y delim char (X-Y grid delimiter between horiz and vertical).

nth 26 = valid wire layer names, "" = all valid.

nth 27,28 = valid wire num, wire num copy layers.

nth 30 = units scl (ex: 1.0 or 25.4).

nth 31 = feature scl (def = 1.0).

nth 45 = Layer for Comp graphis.

nth 46 = Sheet dwg name.

nth 47 = alt xref format.

nth 48 = Fixed tag layer name.

nth 49 = wire gap style.

nth 50 = misc flags:

1's bit = mm fullsize.

- 2's bit = ignore nonlay vector.
- 4's bit = use plc wire numbers.
- 8's bit = fill maxcnt xrefs with "SP".
- 16's bit = srch for PLC address on INS COMP.
- 32+64 bit = 10 = use IEC tee connections, 01 = none, 00 = use DOTS

nth 51-53 = IEC codes (dwg-wide).

nth 54 = ref wireno layer (sig arrow wireno, terminal wireno text layer).

nth 55 = misc data (future).

nth 56 = valid "Fan-in/Fan-out, single-line" layers.

nth 57 = Fan-in/Fan-out symbol style.

nth 58 = Location Box layer.

nth 59 = SORTMODE = re-tag and wire numbering sort mode.

nth 60=XREF_STYLE 0=text, 1=graphical, 2=table

nth 61=XREF_FLAGS= 1'sbit=include unused contacts, 2'sbit=include parent coil (table only)

nth 62=XREF_UNUSEDSTYLE 0=separate reference, 1=contact count totals

nth 63=XREF_FILLWITH fillwith text

nth 64=XREF_SORT 0=sort by line reference, 1=sort by pinlist

nth 65=XREF_TXTBTWN text format. text between separate references

nth 66=XREF_GRAPHIC 0=contact mapping, 1=Graphic

nth 67=XREF_GRAPHICSTYLE 0=JIC, 1=IEC

nth 68=XREF_CONTACTMAP contact mapping list

nth 69=XREF_TBLSTYLE table style name

nth 70=XREF_TBLTITLE table title

nth 71=XREF_TBLINDEX table fields to include

nth 72=XREF_TBLFLDNAMS table available field names

nth 73=XREF_TBLCOLJUST table fields default justification

nth 74=WNUM_OFFSET drawing override for wire placement offset from end of wire, 0.0=centered

nth 75=WNUM_FLAGS

- 1's bit = auto-adjust in-line wire number and label gap width
- 2's bit = wire numbers below wire, 2's bit clear = wire numbers above wire
- 4's bit = in-line wire numbers

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c:ace_ladr2_reseq_main

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Description

Project-wide ladder reference renumber.

AutoLISP

```
(c:ace_ladr2_reseq_main dist ladr_num ladr_mode ladr_skip)
```

Parameters

Parameters	Description
dist	List of drawings to process from active project.
ladr_num	First ladder reference as a string, i.e. "101"
ladr_mode	0 for continuous, 1 for skip (integer)
ladr_skip	Number to skip if ladr_mode is 1. (integer)

Example

Renumber all ladders in active project starting with reference "A100"

```
(setq dlst (nth 4 (c:wd_proj_wdp_data_fnam (ace_getactiveproject))))
(c:ace_ladr2_reseq_main dlst "A100" 0 0)
```

Module

Section E - Drawing Setup

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Example, Section E - Drawing Setup

c:wd_1_reset_dwg

Section E - Drawing Setup Send Feedback

Description

Reset AcadE drawing-specific global variables. For example, when a different drawing is made active.

AutoLISP

```
(c:wd_1_reset_dwg)
```

Module

Section E - Drawing Setup

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Section E - Drawing Setup

What do you think about this topic? Send feedback!

c:wd_chg_ldr_info

Example Section E - Drawing Setup Send Feedback

Description

Retrieve list of ladder parameters from current drawing.

AutoLISP

```
(c:wd_chg_ldr_info oldlst newlst)
```

Parameters

Parameters	Description
oldlst	List as returned by wd_get_gridrefs.
newlst	Edited version of "oldlst" with desired changes.

Example

Bump all ladder references on active drawing up or down by a fixed amount.

```
        (foreach lst oldlst
          ; Current ladder reference text is 2nd element of sublist
          (setq oldref (nth 1 lst))
          ; Increment/decrement by "count"
          (setq newref (c:wd_increment_str oldref count))
          ; Substitute this new value back into the sublist
          (setq lst (wd_1_nth_subst 1 newref lst))
          ; Add to end of new overall list
          (setq newlst (append newlst (list lst)))
       )
        ; Now update this ladder with the new list of sublist data
        (c:wd_chg_ldr_info oldlst newlst)
     )
  )
    (princ) ; suppress writing any return value to command window
 )
```

Module

Section E - Drawing Setup

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Example, Section E - Drawing Setup

c:wd_get_gridrefs

Section E - Drawing Setup Send Feedback

Description

Retrieve list of ladder parameters from current drawing.

AutoLISP

```
(c:wd_get_gridrefs)
```

Returns

List of lists of ladder parameters, one sublist for each ladder on the current drawing. Each sublist is broken down as follows:

nth 0 = XY origin of ladder's first line reference.

nth 1 = Ladder's first line reference text value (attribute RUNGFIRST).

nth 2 = RUNGINC attribute value.

nth 3 = RUNGDIST attribute value.

nth 4 = RUNGCNT attribute value.

nth 5 = entity name of RUNGFIRST attribute.

nth 9 = RUNGHORV code ("H"=horizontal rungs, "V"=vertical rungs).

nth 10 = SHEET attribute value (if present).

nth 11 = entity name of SHEET attribute (if present).

Module

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Section E - Drawing Setup

What do you think about this topic? Send feedback!

AutoCAD Electrical 2025: AutoLISP Reference

c:wd_in_ladder

Example Section E - Drawing Setup Send Feedback

Description

Insert ladder.

AutoLISP

```
(c:wd_in_ladder Lc_ph pt1 pt2 rungl isvert rungw rungv Lc_3sp rungi gridxy_mode Lc_dr def_1st
options)
```

Parameters

Parameters	Description
Lc_ph	1 for single phase ladder or 3 for 3-phase ladder.
pt1	xyz coordinate of left-hand, top-most reference of la
pt2	xyz of bottom ref of ladder. If pt2 = nil then the bott point is calculated as offset from pt1 given in length "rungl."
rungl	Ladder length. (when pt2=nil)
isvert	"V" to create a ladder with vertical rungs (i.e. horiz l or "H" to create ladder with horizontal rungs (i.e. ve ladder). Case sensitive.
rungw	Ladder width.
rungv	Vertical distance between successive rungs.

	ladder that will not have line reference numbers atta to it.
Lc_dr	-1 to suppress drawing all rungs, 0 draw all, 1 to ski every other rung, 2 skip two, draw one.
def_1st	First line ref value (alphanumeric string, not integer)
options	nil or 0 to insert normally with line references and si rails. 1's bit = suppress line reference numbering. 2's bit = suppress side rails.

Example

Insert a vertical ladder (horizontal rungs) at coordinate 58.0, 14.75 that is 20.5 units long, 10.5 wide with 0.75 run spacing, showing line reference numbers beginning with "A100".

```
(c:wd_in_ladder 1 '(58.0 14.75) nil 20.5 "H" 10.5 0.75 0.75 1 nil 0 "A100" nil)
```

The same as above but inserting a horizontal ladder (vertical rungs) and skipping insertion of every other rung.

```
(c:wd_in_ladder 1 '(58.0 14.75) nil 20.5 "V" 10.5 0.75 0.75 1 nil 1 "A100" nil)
```

Module

Section E - Drawing Setup

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AutoCAD Electrical 2025: AutoLISP Reference

c:wd_ladr_reref

Section E - Drawing Setup Send Feedback

Description

Freshen ladder reference numbers on active drawing if they include a SHEET attribute and the WD_M block's SHEET differs from that on each ladder's master line reference block insert.

AutoLISP

```
(c:wd_ladr_reref)
```

Links

Section E - Drawing Setup

c:wd_prnt_time_date

Section E - Drawing Setup Send Feedback

Description

Add current time and date stamp to title block. Annotate title block with current time and date stamp information if attributes are present.

AutoLISP

```
(c:wd_prnt_time_date)
```

Module

Section E - Drawing Setup

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Section E - Drawing Setup

c:wd_read_dwg_params

Section E - Drawing Setup Send Feedback

Description

Read AutoCAD Electrical's WD_M block for active drawing and fill internal global GBL_wd_m. Insert the WD_M block not found on the active drawing (user is prompted for permission to insert).

AutoLISP

```
(c:wd_read_dwg_params)
```

Returns

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AutoCAD Electrical 2025: AutoLISP Reference

c:wd_ref_col

Section E - Drawing Setup Send Feedback

Description

Re-reference ladder reference numbers on active drawing.

AutoLISP

```
(c:wd_ref_col mlr_en first rungi rungv rungc just_erase horv sheet sh_en sub_blk_name)
```

Parameters

Parameters	Description
mlr_en	Entity name of the top-most reference number (the master line reference block insert).
first	First line reference value (alphanumeric string, not integer).
rungi	Line reference rung increment (default = 1).
rungv	Vertical distance between sucessive rungs.
rungc	Count of rung references to update.
just_erase	1 to blank out value on master line ref (the top-most value) and erase existing 2+ line reference numbers
horv	"V" for ladder with vertical rungs, "H" for ladder with horizontal rungs.
sheet	Optional sheet number text value if this is to part of special user hex block (that has its own SHEET attrib
sh_en	SHEET attrib ent name on the special user master lir block insert, nil = none.
sub_blk_name	Name of nested block embedded into special user he block, carries the master line ref attributes.

Links

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What do you think about this topic? Send feedback!

AutoCAD Electrical 2025: AutoLISP Reference

c:wd_renum_ladders

Example Section E - Drawing Setup Send Feedback

Description

Re-reference ladder reference numbers on active drawing with minimum information required.

AutoLISP

```
(c:wd_renum_ladders lst)
```

Parameters

Parameters	Description
lst	list of one or more sublists, each sublist consisting o mimimum of two and max of five pieces of informati old_first_num = existing first line ref number on tarç ladder. new_first_num = replacement for above (three optic parameters, use "nil" if not defined). rungi = optional line ref rung increment (if different existing). rungv = optional rung spacing value (if different fror existing). rungc = optional count of rung refs to update.

Example

Look for ladders beginning with "101" and "120" on the active drawing and, if found, renumber them beginning witl and "320" respectively.

```
(c:wd_renum_ladders (list (list "101" "301")(list "120" "320")))
```

Module

Section E - Drawing Setup

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AutoCAD Electrical 2025: AutoLISP Reference

c:wd_reread_dwg_params

Section E - Drawing Setup Send Feedback

Description

Provided for backwards compatibility. Use wd_read_dwg_params or ace_GBL_wd_m instead.

AutoLISP

```
(c:wd_reread_dwg_params)
```

Module

Section E - Drawing Setup

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AutoCAD Electrical 2025: AutoLISP Reference

c:wd_tb_process_one

Example 1 Example 2 Example 3 Section E - Drawing Setup Send Feedback

Description

Update title block attribute values.

AutoLISP

```
(c:wd_tb_process_one suppress togglelst)
```

Parameters

Parameters	Description
suppress	1 to suppress any error alert boxes, 0 to allow.
togglelst	list of update toggle flags consisting of a 13-element

19). This list of LINEn attribute references can be ze
unlimited length. 1st = mapped to DWGDESC (the description under drawing "properties").
2nd = mapped to DWGSEC.
3rd = mapped to DWGSUB.
4th = mapped to FILENAME.
5th = mapped to FILENAMEEXT.
6th = mapped to FULLFILENAME.
7th = mapped to DWGNAM.
8th = mapped to SHEET.
9th = mapped to SHEETMAX.
10th = (not used for single drawing update)
11th = mapped to IEC_P (the drawing's IEC coding c
for %P).
12th = mapped to IEC_I (the drawing's IEC coding c
for %I).
13th = mapped to IEC_L (the drawing's IEC coding c
for %L).
14th = mapped to 1st line only of DWGDESC
15th = mapped to 2nd line only of DWGDESC
16th = mapped to 3rd line only of DWGDESC

Example 1

Update TB of active dwg, update attribs mapped to LINE1, LINE2, LINE3, and LINE37 as displayed in the UPDATE T dialog box.

```
(c:wd_tb_process_one 0 (list (list 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0) 0 1 2 36))
```

Example 2

Update TB of active dwg, update attribs mapped to three lines of DWGDESC, SHEET, LINE2-3 and LINE12.

```
(c:wd_tb_process_one 0 (list (list 1 0 0 0 0 0 0 1 0 0 0 0 0 1 1 1) 1 2 11))
```

Example 3

Update TB of active dwg, update SHEET and SHEETMAX only.

```
(c:wd_tb_process_one 0 (list (list 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0 0)))
```

Module

Section E - Drawing Setup

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